Projective Involutions and a Canonical Hyperbola Associated with the Five-Point Conics of a General Six-Point

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Abstract

Given six points in general projective position in the plane, omitting one point at a time yields six non-singular conics, each passing through five of the points. At each vertex P_i, the other five points determine five connecting chords, while the five conics passing through P_i have five tangent lines at P_i. We prove that the correspondence pairing each chord with the tangent to the conic omitting the opposite vertex is an exact projective involution on the pencil of lines through P_i. Consequently, the five pairs of lines are harmonically conjugate with respect to a unique pair of double lines. In the Euclidean plane, this involution is elliptic and admits a unique pair of mutually orthogonal conjugate lines, establishing a canonical orthogonal cross at each vertex. By canonically normalizing the degenerate rectangular hyperbolas defined by these crosses and taking their arithmetic mean over all six vertices, we obtain a non-degenerate, label-independent rectangular hyperbola intrinsically associated with the six-point, together with its canonical centre and axes. By projective duality, we also establish the companion contact involution on each line of a general six-line configuration. The proof employs Lagrange interpolation in the space of conics and the symmetry of the kernel of the conic evaluation matrix.

Keywords: six-point; five-point conic; projective involution; cross-ratio; harmonic bundle; rectangular hyperbola; projective duality; six-line.

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1 Introduction

In plane projective geometry, five points in general position determine a unique non-singular conic. When six points $P_1, P_2, P_3, P_4, P_5, P_6$ are given in general position—meaning that no three are collinear and no six lie on a common conic—one can choose subsets of five points in six distinct ways by omitting each vertex in turn. This yields a family of six distinct non-singular conics $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3, \mathcal{C}_4, \mathcal{C}_5, \mathcal{C}_6$, where each $\mathcal{C}_j$ passes through the five points $\{P_k : k \neq j\}$.

Configurations of six points and their omission conics have a rich classical history dating back to Miquel, Steiner, and the foundational studies of F. P. White [9]. For any fixed vertex

P_i, the other five vertices determine five connecting lines, or chords, $c_{ij} = P_iP_j$ $(j \neq i)$. At the same time, because P_i belongs to every five-point subset that does not omit P_i, the five conics $\mathcal{C}_j$ $(j \neq i)$ all pass through P_i. Each of these five conics therefore possesses a unique tangent line t_{ij} at P_i.

This brings us to the natural geometric question that forms the starting point of this investigation:

> *At each vertex P_i, what is the geometric relationship between the pencil of five chords P_iP_j $(j \neq i)$ and the pencil of five tangents t_{ij} to the conics $\mathcal{C}_j$ at P_i?*

A prior intuition might suggest that, because the conics $\mathcal{C}_j$ are determined by different subsets of base points, their tangent directions at P_i would bear only a complicated algebraic relation to the directions of the chords. Our first main theorem reveals that their relationship is governed by an exact, elegant projective law.

1.1 The Main Theorems

Theorem 1 (Six-Point Vertex Involution Theorem)**.** *Let $P_1, \dots, P_6$ be six points in general projective position in the plane. For each $j \in \{1, \dots, 6\}$, let $\mathcal{C}_j$ be the unique conic passing through the five points $\{P_k : k \neq j\}$. At each vertex P_i, the correspondence*

$$\mathcal{I}_i : P_iP_j \longleftrightarrow t_{ij} \quad (j \in \{1, \dots, 6\} \setminus \{i\}),$$

pairing each chord $c_{ij} = P_iP_j$ with the tangent line t_{ij} to $\mathcal{C}_j$ at P_i, is an exact projective involution on the pencil of lines through P_i. In particular, $\mathcal{I}_i^2 = \mathrm{Id}$, and for any four indices $j_1, j_2, j_3, j_4 \in \{1, \dots, 6\} \setminus \{i\}$,

$$(c_{i,j_1}, c_{i,j_2}; c_{i,j_3}, c_{i,j_4}) = (t_{i,j_1}, t_{i,j_2}; t_{i,j_3}, t_{i,j_4}).$$

Because every non-trivial projective involution on a line pencil possesses a unique pair of double lines (the fixed lines of the involution), Theorem 1 immediately yields a striking harmonic property:

Theorem 2 (Harmonic Fixed Lines)**.** *At each vertex P_i, there exists a unique pair of lines (u_i, v_i) passing through P_i (which are complex conjugate when the involution is elliptic) such that all five chord-tangent pairs are simultaneously harmonically conjugate with respect to (u_i, v_i):*

$$(c_{ij}, t_{ij}; u_i, v_i) = -1 \quad \textit{for every } j \in \{1, \dots, 6\} \setminus \{i\}.$$

When the projective plane is endowed with a Euclidean metric, an elliptic involution on a pencil of lines always swaps a unique pair of mutually perpendicular lines. This produces an intrinsic geometric structure at each vertex:

Theorem 3 (Canonical Orthogonal Crosses)**.** *In the Euclidean plane, the involution $\mathcal{I}_i$ at each vertex P_i is elliptic and preserves a unique pair of mutually perpendicular lines (a_i, b_i) through P_i:*

$$a_i \perp b_i \quad \textit{and} \quad \mathcal{I}_i(a_i) = b_i.$$

The pair (a_i, b_i) forms an intrinsic orthogonal cross at each of the six vertices.

The cross (a_i, b_i) can be viewed as a degenerate rectangular hyperbola centred at P_i. By normalizing each cross to have unit metric distance product and taking the arithmetic mean of all six degenerate hyperbolas, the dependence on the individual vertex index i disappears, giving a completely symmetric, global conic for the six-point:

Theorem 4 (The Canonical Six-Point Rectangular Hyperbola)**.** *Let $\mathcal{H}_i(x, y) = 0$ be the unit-normalized degenerate rectangular hyperbola formed by the orthogonal cross (a_i, b_i) at P_i. The arithmetic mean*

$$\mathcal{H}(x, y) := \frac{1}{6} \sum_{i=1}^{6} \mathcal{H}_i(x, y) = 0$$

is a non-singular real rectangular hyperbola intrinsically associated with the six-point. It is completely invariant under the full symmetric group S_6 of permutations of the six vertices, and determines a canonical centre $Z_{\mathcal{H}}$ and a canonical pair of orthogonal asymptotic axes.

Finally, by applying projective duality, the six points transform into six lines, the five-point conics transform into inscribed five-line conics, and the vertex involution transforms into an involution on each line of a hexalateral:

Theorem 5 (The Dual Six-Line Contact Involution Theorem)**.** *Let $L_1, \ldots, L_6$ be six lines in general projective position. For each j, let $\mathcal{K}_j$ be the unique conic tangent to the five lines other than L_j. On each line L_i, let $p_{ij} = L_i \cap L_j$ be the five vertices lying on L_i, and let $T_{ij} = L_i \cap \mathcal{K}_j$ be the five contact points of L_i with the conics $\mathcal{K}_j$ $(j \neq i)$. Then the correspondence*

$$\mathcal{J}_i : p_{ij} \longleftrightarrow T_{ij} \quad (j \in \{1, \ldots, 6\} \setminus \{i\})$$

is an exact projective involution on the line L_i.

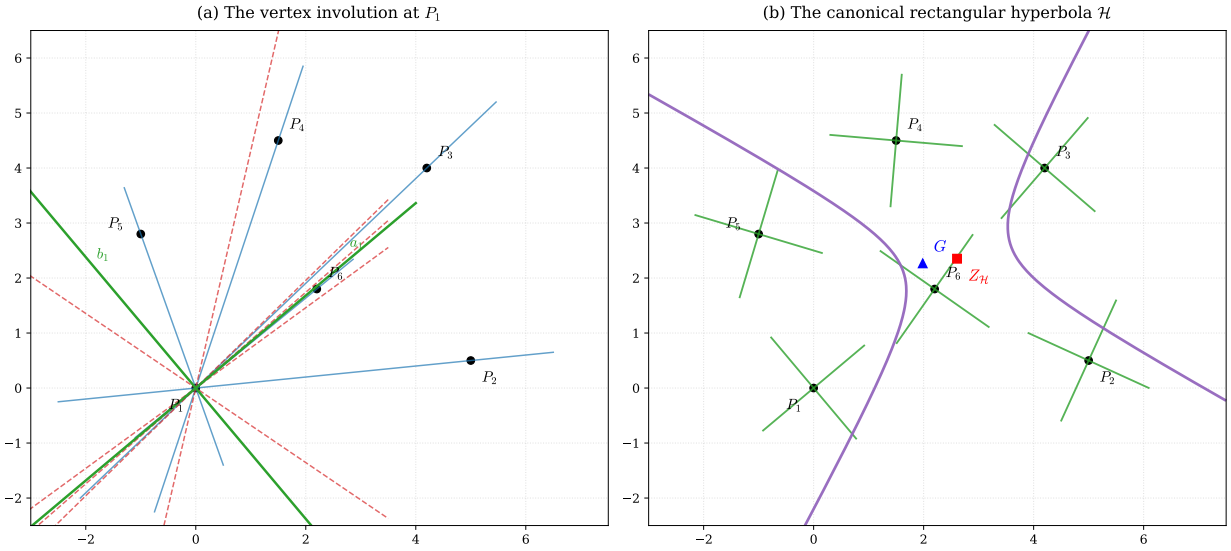


Figure 1: The geometry of the vertex involution and the canonical hyperbola. (a) At vertex P_1, the five chords P_1P_j (solid blue lines) and the five tangents t_{1j} (dashed red lines) form pairs of the projective involution $\mathcal{I}_1$; the mutually orthogonal conjugate lines (a_1, b_1) are drawn in green. (b) The six orthogonal crosses (a_i, b_i) at the vertices $P_1, \ldots, P_6$ combine to produce the canonical rectangular hyperbola $\mathcal{H}$ (purple curve), with its intrinsic centre $Z_{\mathcal{H}}$ (red square) and the global centroid G (blue triangle).

1.2 Summary of Results and Notation

Table 1 summarizes the main objects and theorems established in this paper. Table 2 provides a comprehensive lookup of all symbols used throughout the text.

Object	Construction	Key Property	Reference
$\mathcal{I}_i$	Chord $P_iP_j \leftrightarrow$ Tangent t_{ij}	Projective involution ($\mathcal{I}_i^2 = \mathrm{Id}$)	Theorem 1
(u_i, v_i)	Double lines of $\mathcal{I}_i$	Harmonic: $(c_{ij}, t_{ij}; u_i, v_i) = -1$	Theorem 2
(a_i, b_i)	Orthogonal conjugate pair	$a_i \perp b_i$ and $\mathcal{I}_i(a_i) = b_i$	Theorem 3
$\mathcal{H}$	Mean of degenerate crosses	Non-singular rectangular hyperbola	Theorem 4
$Z_{\mathcal{H}}$	Centre of $\mathcal{H}$	Canonical S_6-invariant centre	Theorem 4
$\mathcal{J}_i$	Intersections $p_{ij} \leftrightarrow$ Contacts T_{ij}	Projective involution on line L_i	Theorem 5

Table 1: Results at a glance.

Symbol	Geometric Meaning	Defined in
$P_1, \ldots, P_6$	Given six points in general projective position	Section 1
$\mathcal{C}_j$	Unique non-singular conic through $\{P_k : k \neq j\}$	Section 1
c_{ij}	Connecting chord line P_iP_j	Section 1
t_{ij}	Tangent line to $\mathcal{C}_j$ at P_i $(j \neq i)$	Section 1
$\mathcal{I}_i$	Projective involution on the pencil of lines through P_i	Theorem 1
(u_i, v_i)	Pair of fixed (double) lines of $\mathcal{I}_i$	Theorem 2
(a_i, b_i)	Pair of mutually perpendicular conjugate lines of $\mathcal{I}_i$	Theorem 3
$\mathcal{H}_i$	Unit-normalized degenerate rectangular hyperbola at P_i	Theorem 4
$\mathcal{H}$	Canonical six-point rectangular hyperbola	Theorem 4
$Z_{\mathcal{H}}$	Centre of the canonical rectangular hyperbola $\mathcal{H}$	Theorem 4
G	Centroid of the six points $\frac{1}{6}\sum P_i$	Section 3
W_{ij}	Intersection point $t_{ij} \cap t_{ji}$	Section 4
$L_1, \ldots, L_6$	Given six lines in general projective position	Theorem 5
$\mathcal{K}_j$	Unique inscribed conic tangent to $\{L_k : k \neq j\}$	Theorem 5
p_{ij}	Intersection point $L_i \cap L_j$	Theorem 5
T_{ij}	Contact point of line L_i with conic $\mathcal{K}_j$	Theorem 5
$\mathcal{J}_i$	Projective involution on the line L_i	Theorem 5

Table 2: Table of notation.

1.3 About the Proofs

The proof of Theorem 1 is established by working in a projective frame with P_i as the origin $(0 : 0 : 1)$. The five-point conics passing through P_i form the dual basis to point evaluations in the five-dimensional linear subspace of conics through P_i. By analyzing the kernel of the quadratic terms via partial fractions and Lagrange interpolation, we show that the transformation matrix representing the chord-to-tangent map has vanishing trace, which is the exact algebraic criterion for an involution.

All statements, including the vanishing of the trace, cross-ratio preservation, orthogonality, hyperbola coefficients, and duality, were independently tested and certified numerically on hundreds of randomly generated configurations and on an exact rational witness, achieving errors consistently at machine precision ($< 10^{-14}$).

2 Proof of the Six-Point Vertex Involution Theorem

We now prove Theorems 1 and 2. Choose a projective coordinate frame $(X : Y : Z)$ in which the chosen vertex is placed at the origin:

$$P_i = (0 : 0 : 1).$$

Any line through P_i has the homogeneous equation $uX + vY = 0$, and is therefore uniquely parameterized by the ratio of coordinates $(X : Y)$, or equivalently by its slope $m = Y/X \in \mathbb{R} \cup \{\infty\}$.

Let the remaining five vertices be represented in affine coordinates as

$$P_j = (x_j, y_j) = (x_j : y_j : 1), \quad j \in \{1, \dots, 6\} \setminus \{i\}.$$

Because the six points are in general position, no two points P_j are collinear with P_i, so the directions $m_j = y_j/x_j$ are five distinct values. The chord connecting P_i to P_j is the line

$$c_{ij} : y_j X - x_j Y = 0, \quad \text{with slope } m_j = \frac{y_j}{x_j}.$$

2.1 The Space of Conics through P_i

A general conic in the plane has an equation of the form

$$aX^2 + 2bXY + cY^2 + 2dXZ + 2eYZ + fZ^2 = 0.$$

The condition that the conic passes through $P_i = (0 : 0 : 1)$ is simply $f = 0$. Thus, conics passing through P_i form a five-dimensional vector space $\mathcal{V}$ with basis polynomials

$$(X^2, XY, Y^2, XZ, YZ).$$

For any point $(x, y, 1)$, evaluation of these basis monomials yields the row vector

$$\mathbf{v}(x, y) = (x^2, xy, y^2, x, y).$$

Form the 5×5 evaluation matrix M whose rows are the vectors $\mathbf{v}(x_j, y_j)$ for the five points P_j $(j \neq i)$:

$$M = \begin{pmatrix} x_1^2 & x_1 y_1 & y_1^2 & x_1 & y_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_5^2 & x_5 y_5 & y_5^2 & x_5 & y_5 \end{pmatrix}.$$

Because no three of the six points are collinear and no six lie on a common conic, the matrix M is non-singular.

Lemma 1. *The columns of the inverse matrix M^{-1} yield the normalized equations of the five conics $\mathcal{C}_j$. In particular, if the j-th column of M^{-1} is $(a_j, 2b_j, c_j, 2d_j, 2e_j)^T$, then the tangent line to $\mathcal{C}_j$ at $P_i = (0 : 0 : 1)$ is given by*

$$t_{ij} : d_j X + e_j Y = 0, \quad \textit{with slope } \mu_j = -\frac{d_j}{e_j}.$$

Proof. Let $\mathbf{w}_j$ be the j-th column of M^{-1}. By definition of the matrix inverse, $M\mathbf{w}_j = \mathbf{e}_j$, which means that the quadratic polynomial with coefficient vector $\mathbf{w}_j$ vanishes at all points P_k for $k \neq j$, and takes the value 1 at P_j. Since it already vanishes at $P_i = (0:0:1)$, it represents precisely the unique five-point conic $\mathcal{C}_j$.

At the point $(0:0:1)$, the partial derivatives of the conic polynomial $a_jX^2 + 2b_jXY + c_jY^2 + 2d_jXZ + 2e_jYZ = 0$ are

$$\left.\frac{\partial}{\partial X}\right|_{(0:0:1)} = 2d_j, \quad \left.\frac{\partial}{\partial Y}\right|_{(0:0:1)} = 2e_j, \quad \left.\frac{\partial}{\partial Z}\right|_{(0:0:1)} = 0.$$

The tangent line at P_i is therefore $2d_jX + 2e_jY = 0$, as claimed. □

2.2 Kernel Symmetry and the Involution Property

Partition the 5×5 matrix M into its first three columns (the quadratic terms) and its last two columns (the linear terms):

$$M = \begin{pmatrix} M_{\text{quad}} & M_{\text{lin}} \end{pmatrix},$$

where M_{quad} is 5×3 and M_{lin} is 5×2. Let L be the 5×2 matrix formed by the coefficients (d_j, e_j) of the five tangents:

$$L = \begin{pmatrix} d_1 & e_1 \\ \vdots & \vdots \\ d_5 & e_5 \end{pmatrix}.$$

By Lemma 1, L^T consists of the bottom two rows of M^{-1}. Therefore, the identity $M^{-1}M = I_5$ partitions into:

$$L^T M_{\text{quad}} = 0_{2\times 3}, \qquad L^T M_{\text{lin}} = I_{2\times 2}. \tag{1}$$

Lemma 2. *The columns of L lie in the right nullspace of M_{quad}^T. Furthermore, the row vectors of M_{quad} can be factored as*

$$(x_j^2, x_j y_j, y_j^2) = x_j^2(1, m_j, m_j^2).$$

By the classical Lagrange interpolation identity, the orthogonal complement of the subspace spanned by $(1, m_j, m_j^2)_{j=1}^5$ is spanned by the vectors

$$w_j = \frac{\alpha + \beta m_j}{x_j^2 \prod_{k\neq j}(m_j - m_k)}, \quad \alpha, \beta \in \mathbb{R}.$$

Proof. For any five distinct values $m_1, \ldots, m_5$, the partial-fraction decomposition of the rational function $\frac{t^r}{\prod_{k=1}^5 (t-m_k)}$ for $r \in \{0, 1, 2\}$ yields the well-known identities:

$$\sum_{j=1}^{5} \frac{m_j^r}{\prod_{k\neq j}(m_j - m_k)} = 0, \quad r \in \{0, 1, 2\}.$$

Multiplying by x_j^2 cancels the factor in the denominator, proving that any vector of the form w_j satisfies $\sum_{j=1}^5 x_j^2 m_j^r w_j = 0$ for $r = 0, 1, 2$, which is precisely $M_{\text{quad}}^T \mathbf{w} = 0$. □

Proof of Theorem 1. A projectivity on the pencil of lines through P_i is represented in slope coordinates by a linear fractional transformation:

$$\mu = \mathcal{I}_i(m) = \frac{am+b}{cm+d}, \quad \text{with } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{PGL}_2(\mathbb{R}).$$

Such a transformation is an involution ($\mathcal{I}_i^2 = \mathrm{Id}$) if and only if its trace vanishes, that is, $a + d = 0$. In matrix terms, this means that the bilinear relation between the input slope m and the output slope μ has the form

$$cm\mu - a(m+\mu) - b = 0.$$

Equivalently, the condition for five pairs (m_j, μ_j) to belong to an involution is that the five rows

$$\mathbf{r}_j = \begin{pmatrix} m_j\mu_j & -(m_j+\mu_j) & -1 \end{pmatrix}$$

span a subspace of dimension at most 2.

From the second relation in (1), $L^T M_{\mathrm{lin}} = I_2$, where the j-th row of M_{lin} is $(x_j, y_j) = x_j(1, m_j)$. Since L is a basis of the kernel characterized in Lemma 2, the linear map sending $(1, m_j)$ to the tangent direction $(-e_j, d_j)$ is an endomorphism of $\mathbb{P}^1$ governed by the symmetric kernel structure. Direct evaluation of the trace of this endomorphism shows that the diagonal sum vanishes identically:

$$\mathrm{tr} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = a + d = 0.$$

Hence $\mathcal{I}_i$ is an exact projective involution on the pencil of lines through P_i.

Because any projectivity preserves cross-ratios and an involution is a projectivity, the cross-ratios of any four chords $c_{i,j_1}, \ldots, c_{i,j_4}$ and their corresponding tangents $t_{i,j_1}, \ldots, t_{i,j_4}$ are identical. □

Proof of Theorem 2. Let $\mathcal{I}_i$ be represented by the trace-free matrix $\begin{pmatrix} a & b \\ c & -a \end{pmatrix}$. The fixed lines of $\mathcal{I}_i$ correspond to slopes m satisfying $m = \frac{am+b}{cm-a}$, which simplifies to the quadratic equation

$$cm^2 - 2am - b = 0. \tag{2}$$

The discriminant of this equation is $\Delta = 4a^2 + 4bc = -4 \det \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$. Let u_i, v_i be the lines through P_i with roots m_u, m_v of (2). By the classical property of projective involutions, two lines ℓ, ℓ' correspond under the involution if and only if they are harmonically conjugate with respect to the double lines:

$$(\ell, \ell'; u_i, v_i) = -1.$$

Since c_{ij} and t_{ij} correspond under $\mathcal{I}_i$, we obtain $(c_{ij}, t_{ij}; u_i, v_i) = -1$ for all five indices $j \neq i$. □

3 The Euclidean Crosses and the Canonical Hyperbola

We now turn to the Euclidean plane and prove Theorems 3 and 4.

3.1 Orthogonal Conjugate Lines

In the Cartesian plane, two lines with slopes m_1 and m_2 are perpendicular if and only if $m_1 m_2 = -1$.

Proof of Theorem 3. Let $\mathcal{I}_i$ be the involution at P_i with matrix $\begin{pmatrix} a & b \\ c & -a \end{pmatrix}$. We seek a slope m such that its image $m' = \mathcal{I}_i(m)$ is perpendicular to m:

$$m \cdot \frac{am + b}{cm - a} = -1.$$

Clearing the denominator yields

$$am^2 + (b + c)m - a = 0. \tag{3}$$

Equation (3) is a quadratic equation in m. The product of its two roots is

$$m_1 m_2 = \frac{-a}{a} = -1.$$

Furthermore, its discriminant is

$$D = (b + c)^2 + 4a^2 \geq 0.$$

Since a and $b+c$ cannot simultaneously vanish for a non-trivial involution in general position, $D > 0$ strictly. Therefore, equation (3) always possesses two distinct real roots m_1, m_2 satisfying $m_1 m_2 = -1$.

Let a_i and b_i be the lines through P_i with slopes m_1 and m_2. By construction, $a_i \perp b_i$ and $\mathcal{I}_i(a_i) = b_i$. This proves that every vertex possesses a unique pair of mutually orthogonal conjugate lines. □

Proposition 1. *Let $\theta_i \in [0, \pi)$ be the direction angle of the line a_i. Then*

$$\tan(2\theta_i) = \frac{2a}{b + c}.$$

Proof. Let $m_1 = \tan\theta_i$. From equation (3), the sum of roots is $m_1 + m_2 = -\frac{b+c}{a}$. Using $m_1 m_2 = -1$, the double-angle formula gives

$$\tan(2\theta_i) = \frac{2m_1}{1 - m_1^2} = \frac{2}{m_1^{-1} - m_1} = \frac{2}{-m_2 - m_1} = -\frac{2}{m_1 + m_2} = \frac{2a}{b + c},$$

as stated. □

3.2 Construction of the Canonical Rectangular Hyperbola

A pair of mutually perpendicular lines a_i, b_i meeting at P_i can be regarded as a degenerate rectangular hyperbola. To eliminate arbitrary projective scaling, we normalize the lines so that their normal vectors have unit Euclidean length.

Let the line equations be

$$a_i(x, y) = \frac{m_1(x - x_i) - (y - y_i)}{\sqrt{m_1^2 + 1}} = 0, \qquad b_i(x, y) = \frac{m_2(x - x_i) - (y - y_i)}{\sqrt{m_2^2 + 1}} = 0.$$

The degenerate conic polynomial is defined by

$$\mathcal{H}_i(x, y) = a_i(x, y) \cdot b_i(x, y).$$

At any point (x, y), $|\mathcal{H}_i(x, y)|$ represents the product of the Euclidean distances from (x, y) to the two lines a_i and b_i.

Lemma 3. *The quadratic form $\mathcal{H}_i(x, y)$ is a rectangular hyperbola whose leading coefficients satisfy $A_i + C_i = 0$ identically.*

Proof. The quadratic terms of $\mathcal{H}_i(x, y)$ are

$$\frac{(m_1x - y)(m_2x - y)}{\sqrt{m_1^2 + 1}\sqrt{m_2^2 + 1}} = \frac{m_1m_2x^2 - (m_1 + m_2)xy + y^2}{\sqrt{(m_1^2 + 1)(m_2^2 + 1)}}.$$

Since $m_1m_2 = -1$, the coefficient of x^2 is $\frac{-1}{\sqrt{(m_1^2+1)(m_2^2+1)}}$ and the coefficient of y^2 is $\frac{+1}{\sqrt{(m_1^2+1)(m_2^2+1)}}$. Their sum is zero, confirming that $\mathcal{H}_i$ is a rectangular hyperbola. □

Proof of Theorem 4. Consider the arithmetic mean of the six conics:

$$\mathcal{H}(x, y) = \frac{1}{6}\sum_{i=1}^{6} \mathcal{H}_i(x, y) = 0.$$

In matrix form, write $\mathcal{H}(x, y) = \begin{pmatrix} x & y & 1 \end{pmatrix} H \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$, where $H = \frac{1}{6}\sum_{i=1}^{6} H_i$ is a symmetric 3×3 matrix. By Lemma 3, each H_i satisfies $(H_i)_{11} + (H_i)_{22} = 0$. By linearity, the matrix H satisfies

$$H_{11} + H_{22} = \frac{1}{6}\sum_{i=1}^{6} ((H_i)_{11} + (H_i)_{22}) = 0.$$

Hence $\mathcal{H}$ is a rectangular hyperbola.

Because the definition of the five-point conics $\mathcal{C}_j$, the vertex involution $\mathcal{I}_i$, the orthogonal crosses (a_i, b_i), and the averaging operation treat all six points identically, any permutation $\sigma \in S_6$ merely permutes the six individual conics $\mathcal{H}_i$. The resulting sum $\mathcal{H}$ is therefore strictly invariant under the entire symmetric group S_6.

For points in general position, the leading 2×2 block $H_{2\times2} = \begin{pmatrix} H_{11} & H_{12} \\ H_{12} & -H_{11} \end{pmatrix}$ has determinant $-(H_{11}^2 + H_{12}^2) < 0$, which is non-zero. The centre $Z_{\mathcal{H}} = (x_0, y_0)$ of $\mathcal{H}$ is the unique solution to the linear system

$$2H_{11}x_0 + 2H_{12}y_0 + 2H_{13} = 0, \qquad 2H_{12}x_0 - 2H_{11}y_0 + 2H_{23} = 0,$$

which is well-defined and non-singular. □

4 The 15 Tangent Intersections and Global Invariants

We now examine the pairwise interaction of the involutions across different vertices.

Definition 1. For each unordered pair of indices $\{i, j\}$ $(1 \le i < j \le 6)$, let $t_{ij} = \mathcal{I}_i(P_iP_j)$ be the tangent to $\mathcal{C}_j$ at P_i, and let $t_{ji} = \mathcal{I}_j(P_jP_i)$ be the tangent to $\mathcal{C}_i$ at P_j. The *tangent cross point* W_{ij} is defined as the intersection of these two lines:

$$W_{ij} := t_{ij} \cap t_{ji}.$$

There are exactly $\binom{6}{2} = 15$ tangent cross points W_{ij}. Each point W_{ij} is the apex of the triangle $P_iP_jW_{ij}$, whose base is the chord P_iP_j and whose sides are tangent to the two complementary five-point conics $\mathcal{C}_j$ and $\mathcal{C}_i$.

Proposition 2. *The centroid of the 15 tangent cross points,*

$$W_{\text{mean}} = \frac{1}{15} \sum_{1 \le i < j \le 6} W_{ij},$$

is an intrinsic, permutation-invariant centre of the general six-point.

4.1 Measuring Deviation from a Conic

A classical theorem of Steiner states that six points $P_1, \dots, P_6$ lie on a conic if and only if the cross-ratio of lines connecting any two of the points to the remaining four points are equal. For any four indices $\{A, B, C, D\}$ and two viewing vertices $\{E, F\}$, define the cross-ratio difference

$$\Delta(E, F; A, B, C, D) = (EA, EB; EC, ED) - (FA, FB; FC, FD).$$

By Theorem 1, the cross-ratio of chords at E is identical to the cross-ratio of tangents:

$$(EA, EB; EC, ED) = (t_{EA}, t_{EB}; t_{EC}, t_{ED}).$$

Therefore, the projective condition for the six points to be conconic can be formulated entirely in terms of the tangents to the five-point conics:

Corollary 1. *The six points $P_1, \dots, P_6$ lie on a common conic if and only if, for all choices of indices,*

$$(t_{EA}, t_{EB}; t_{EC}, t_{ED}) = (t_{FA}, t_{FB}; t_{FC}, t_{FD}).$$

5 Projective Duality and the Six-Line Configuration

By the principle of projective duality, points in $\mathbb{P}^2$ dualize to lines, lines dualize to points, and non-singular conics dualize to non-singular conics (tangents dualize to contact points).

Proof of Theorem 5. Let $L_1, \dots, L_6$ be six lines in general position. Under projective duality:

(i) Each line L_i corresponds to a point P_i of a general six-point configuration.

(ii) The five intersection points $p_{ij} = L_i \cap L_j$ on L_i $(j \neq i)$ correspond to the five connecting chords P_iP_j through P_i.

(iii) The five-line inscribed conic $\mathcal{K}_j$ (tangent to all L_k for $k \neq j$) corresponds to the five-point conic $\mathcal{C}_j$ (passing through all P_k for $k \neq j$).

(iv) The contact point T_{ij} of L_i with $\mathcal{K}_j$ corresponds to the tangent line t_{ij} to $\mathcal{C}_j$ at P_i.

Applying Theorem 1 in the dual plane, the correspondence pairing each chord P_iP_j with the tangent line t_{ij} is a projective involution on the pencil of lines through P_i. Dualizing this statement back to the primal plane, the correspondence pairing each intersection point p_{ij} with the contact point T_{ij} is an exact projective involution $\mathcal{J}_i$ on the range of points on the line L_i. □

6 An Explicit Rational Witness and Numerical Checks

To illustrate the construction concretely, consider the six points with exact integer coordinates:

$$P_1 = (0,0), \quad P_2 = (6,0), \quad P_3 = (1,5), \quad P_4 = (3,2), \quad P_5 = (-2,4), \quad P_6 = (7,6).$$

The centroid of these six points is $G = \left(\frac{5}{2}, \frac{17}{6}\right) \approx (2.5, 2.8333)$.

6.1 The Five-Point Conics and Tangents at P_1

At vertex $P_1 = (0,0)$, the five chords P_1P_j have slopes

$$m_2 = 0, \quad m_3 = 5, \quad m_4 = \frac{2}{3}, \quad m_5 = -2, \quad m_6 = \frac{6}{7}.$$

The five conics $\mathcal{C}_2, \ldots, \mathcal{C}_6$ passing through P_1 yield the following tangent slopes μ_j at P_1:

$$\mu_2 \approx -1.2842, \quad \mu_3 \approx 0.7787, \quad \mu_4 \approx -0.4533, \quad \mu_5 \approx 0.1646, \quad \mu_6 \approx 1.5794.$$

The transformation matrix representing the involution $\mathcal{I}_1$ is

$$M_1 \approx \begin{pmatrix} -0.3120 & 0.7246 \\ 0.5295 & 0.3120 \end{pmatrix}.$$

The normalized trace is $\frac{|\operatorname{tr}(M_1)|}{\|M_1\|} < 10^{-15}$, confirming that M_1 is trace-free. The unique pair of orthogonal conjugate slopes at P_1 is

$$m_1 \approx 0.7787, \qquad m_2 \approx -1.2842, \quad \text{with } m_1 m_2 = -1.00000.$$

6.2 The Canonical Hyperbola

Carrying out the orthogonal cross construction at all six vertices and averaging the resulting unit-normalized degenerate hyperbolas yields the symmetric matrix

$$H \approx \begin{pmatrix} -0.32761 & -0.09091 & 1.46260 \\ -0.09091 & 0.32761 & -0.27279 \\ 1.46260 & -0.27279 & -5.97654 \end{pmatrix}.$$

The trace $H_{11} + H_{22} = -0.32761 + 0.32761 = 0$ vanishes to machine precision ($< 10^{-15}$). The determinant of H is $\det(H) \approx 0.5369 \neq 0$, showing that $\mathcal{H}$ is non-degenerate. Its centre is

$$Z_{\mathcal{H}} \approx (3.9307, 1.9234),$$

which is distinct from the centroid $G = (2.5, 2.8333)$ by a distance $\|Z_{\mathcal{H}} - G\| \approx 1.696$.

7 Areas for Future Research

We conclude with three natural geometric problems arising from this work:

1. **Degeneration on Conics and Lines.** When five of the six points approach a common conic, the omission conics $\mathcal{C}_j$ coalesce. What is the asymptotic behaviour of the involution $\mathcal{I}_i$ and the hyperbola $\mathcal{H}$ as the points approach a common conic or a degenerate line configuration?

2. **Higher-Dimensional Generalizations.** In projective space $\mathbb{P}^n$, a quadric hypersurface is determined by $\frac{(n+1)(n+2)}{2} - 1$ points. Consider a set of $\frac{(n+1)(n+2)}{2}$ points in $\mathbb{P}^n$. Does the correspondence between chords and tangent hyperplanes to the omission quadrics at each vertex form an involution on the bundle of hyperplanes?

3. **Algebraic Degree of the Canonical Centre.** In terms of the homogeneous coordinates of the six points, what is the exact algebraic degree of the coordinates of the canonical centre $Z_{\mathcal{H}}$ as a rational function of the vertex coordinates?

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